AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TT284
Web Technologies

Final Examination
Summer Semester 2025/2026

Time Allowed: 3 hours

Instructions:

- Total Marks: 100 **marks**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Basic Calculators** are allowed.

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PART 1: (True/False questions [20 Marks])
All questions are required (10 questions, 2 marks each one)

1.	Pixel size refers to the physical dimensions of the screen typically measured on the diagonal.
2.	A phone with a stand-alone GPS module allows it to work with the phone network to give a quicker fix than a built-in GPS, and a more accurate one than the phone network or Wifi triangulation alone
3.	Once an app is developed, the next stage is to make it available either within an organization or to the general public. If they are for the general public, a choice has to be made as to whether to charge or not. The process of making them available to the market is called: the 'route to market'.
4.	If a system has not yet been built, the normal calculation for availability cannot be applied, so we have to turn to a slightly different notion of availability based on the mean time to failure (MTTF) and the 'mean time to repair' (MTTR), also known as 'mean time to replace'.
5.	The gains of clustering come with additional costs such as dual network interfaces in each node, one for application data and one for the heartbeat.
6.	Any web application has an infinite life during which a unique set of skills, techniques, tools and resources are brought together to achieve a specific set of goals; they are what we term projects.
7.	The International Organisation for Standardisation (ISO) has published standards relating to software quality such as functional suitability.
8.	Regression testing aims to ensure that the application functions correctly regardless of browser, operating system, or hardware type or version.
9.	One of the most common vulnerabilities associated with application software, such as web, application and database servers, is that users attempt to use the software 'straight from the box' without understanding the implications of what they have installed.
10.	Asymmetric encryption methods use a single key to encrypt and decrypt messages and the computation is fast. The major disadvantage is that the key has to be shared between the two users before messages can be encrypted and exchanged.

PART 2: (Essay questions [30 Marks])
All questions are required (6 questions, 5 marks each one)

1) Modern mobile devices integrate a range of hardware components and network capabilities. Identify four key features that collectively distinguish modern mobile devices from earlier generations of portable technology.

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2) State the different complementary techniques that enable modern mobile devices to determine their geographical location.

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3) Explain how the load sharing technique can be used to increase system availability and describe the key components and structure of a typical load-sharing configuration.

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4) Once project risks have been identified and assessed, several mitigation strategies can be applied to manage them. Explain five of these main risk-handling strategies available to a project manager.

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5) Explain the purpose of a stress test and describe how it helps maintain system reliability and stability when failures such as interrupted user transactions occur under extreme load.

6) Modern web applications face several security risks that can compromise data integrity, user privacy, and system reliability. Identify five of the major categories of these security risks.

PART 3: (Problem Solving Questions [50 Marks])
All questions are required (5 questions, 10 marks each one)

1) Samsung Galaxy S has screen size of 800 x 480 pixels and LG 0ptimus has a screen size of 1024 x 768 pixels. Compare the resolutions in pixels per inch of the Samsung Galaxy S and LG 0ptimus using figure 1 below:

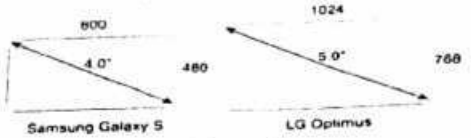


Figure 1.

2) If the mean time to failure of a system is 3,000 hours and the mean time to repair is 300 hours, calculate the inherent availability of the system.

3) What is the output of the following code?

```
<html> <body> <script>
function increase(A, B) {
   A = A + (B % 3);
   return A; }
function decrease(A, B) {
   A = A - (B - 3);
   return A; }
function processData() {
   A = 10;    B = 7;    C = 2;
   for (i = 1; i <= 5; i++) {
      if (i % 3 == 0)
         A = decrease(A, B);
      else
         A = increase(A, B);
      if (A > 12) {
         document.write((A - C) + "<br>");
         B = B - 1; }
      else if (A >= 8) {
         document.write((A + B + C) + "<br>");
         C = C + 2; }
      else {
         document.write((A * 2) + "<br>");   }
      if (C > 6)
         C = C - 4;  } }
processData();
</script>  </body>  </html>
```

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4) Write a PHP script that will perform the following
   a) Creates two associative arrays named *Attempt1* and *Attempt2* with the following values:
      Attempt1: Task_A → 5, Task_B → 2, Task_C → 8
      Attempt2: Task_A → 1, Task_B → 7, Task_C → 4
   b) Calculate and print the average value for each attempt
   c) Find and print the minimum for each task by comparing between the tasks of Attempt1 with the corresponding tasks of Attempt2.

5) Write the HTML code that creates and displays a simple web page containing a form, based on the layout shown in Figure 2 below:

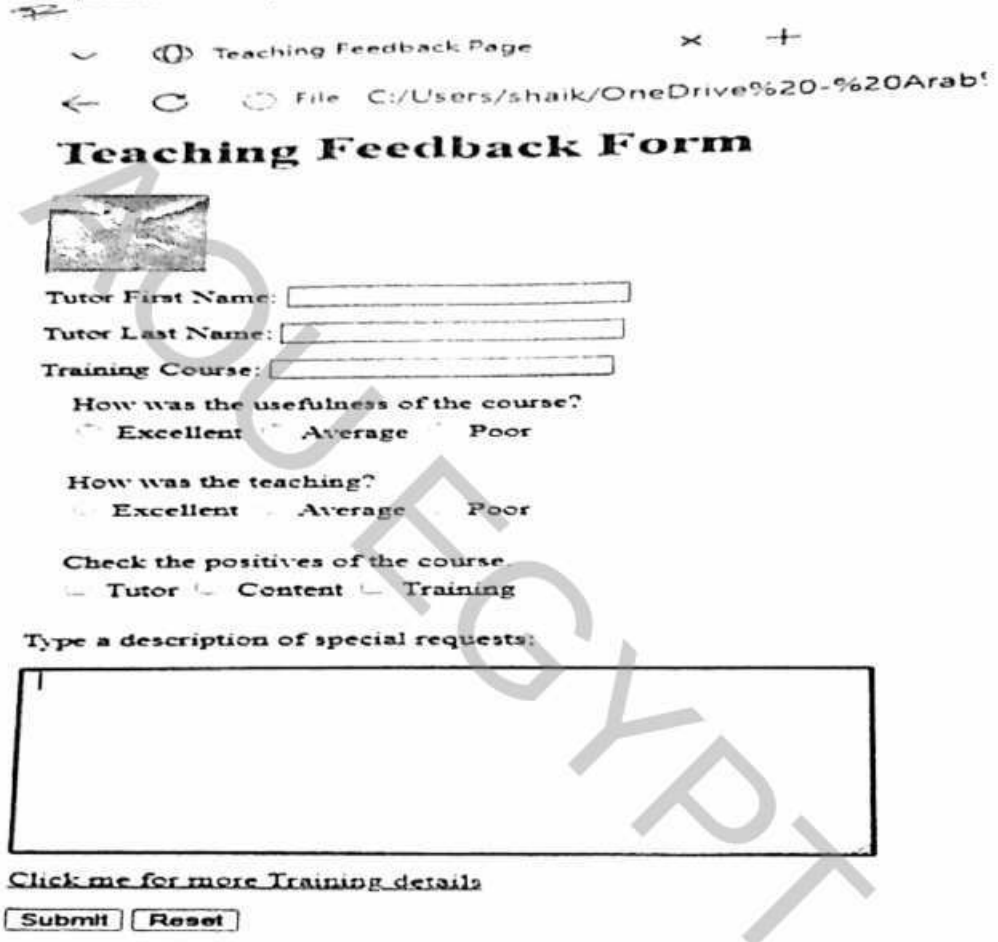


Figure 2

End of questions

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